

Index

S. No.	Name of the Experiment	Page No.	Date of Experiment	Date of Submission	Remarks
01.	To identify and classify different types of cyber threats	01 - 05	03-01-26		
02.	To identify and analyze Signs of potential Security incidents using Sysinternal suite.	06 - 08	22-01-26		
03.	To install & identify, configure Wazuh for monitoring an endpoint.	09- 11	12-02-26		
04.	To perform live incident Response & data acquisition on a windows System using Helix Tool.	12 - 14	26-02-26		
05.	Installation & Demonstration of Sawmill on windows OS & Generation of a custom Report.	15 - 18	05-03-26		
06.	To monitor disk activity shared and identify processes where the disk read & write queue length exceeds 5 using windows monitoring tools.	19 - 21	26-03-26		

Index

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01. AIM: To analyze and classify different types of cyber threat incidents and prepare a strategic method of action to identify, prevent, and mitigate the incident as an incident Response importance.

02. OBJECTIVES:

1. To classify major cyber threats.
2. To identify Indicators of compromise for each incident.
3. To analyze attack methods & impacts.
4. To develop suitable prevention and mitigation strategies.
5. To prioritize incident Response based on importance.

03. METHODOLOGY:

The practical was conducted by analyzing five real-world cyber threat incidents, focusing on attack vectors, Indicators of compromise (Iocs), impact, and Response Strategy.

01. Ransomware Attack - Collins Aerospace (VMUSE)

a. Introduction: The Collins Aerospace cyberattack is an attack on VMUSE check-in and boarding software created by Collins Aerospace. The attack affects several European airports. Airports are connected via a platform called VMUSE, which in turn uses a dedicated VPN called ARINC Avinet. ARINC was previously an independent company, but it operated as part of Collins from 2018 until the date of the incident. According to UpGuard, a cybersecurity company, the outage across airports indicates that Avinet was compromised.

b. Attack Analysis: Ransomware attack targeting VMUSE airport check-in and boarding Software via a compromised dedicated VPN (CARINC AvinNet), disrupting airport operations across Europe.

c. Indicator Of Compromise (IOC):

- Sudden outage of check-in and baggage handle systems.
- Manual airport operations and widespread service delays.
- Unusual VPN Traffic and unauthorized access logs.
- Detection of known ransomware malware strain.

d. Assessment: High-impact attack on critical aviation infrastructure requiring immediate containment and recovery.

02. Phishing & Data Breach Campaign - ShinyHunters.

a. Introduction: ShinyHunters is a black hat criminal hacker and extortion group that is believed to have formed in 2020 and is said to have been involved in massively significant amounts of data breaches. The group often extorts the company they've hacked. If the company doesn't pay the ransom, the stolen information is often sold or leaked on the dark web.

b. Attack Analysis: Large-scale phishing and credential theft campaigns leading to unauthorized access, data exfiltration, and extortion of multiple global organizations including Microsoft, Google, AT & T wireless, Aditya Birla Fashion and retail and much more.

c. Indicator of Compromise (IOCs):

- Phishing emails impersonating trusted sources.
- Credential reuse alerts and abnormal login locations.
- Unauthorized database access and large data exports.
- Appearance of leaked data on dark web forums.

d. Assessment: Persistent and financially motivated threats with severe data confidentiality impact.

03. Virus/Worm Incident - WannaCry Ransomware.

a. Introduction: WannaCry is a ransomware crypto worm, which targets computers running the Microsoft Windows operating system by encrypting (locking) data and demanding ransom payments in the Bitcoin cryptocurrency. The worm is also known as Wannacrypt, Wana Decryptor 2.0, Wannacryptor 2.0, and Wanna Decryptor. It is considered a network worm because it also includes a transport mechanism to automatically spread itself.

b. Attack Analysis: Self-propagating ransomware worm exploiting the SMB vulnerability (EternalBlue) and installing the DoublePulsar backdoor for rapid lateral movement.

c. Indicators of Compromise (Iocs):

- Encrypted files with .WNCRY Extension.
- Ransom notes demanding Bitcoin Payment.
- Unexpected SMB Traffic on Port 445.
- Presence of DoublePulsar backdoor artifacts.

d. Assessment: Highly contagious network-wide threat exploiting unpatched systems.

04. Spyware Campaign - PhoneSpy (Android)

a. Introduction: In 2021, Security researchers identified an extensive, ongoing spyware campaign dubbed "PhoneSpy" that targeted thousands of South Korean Android users. This was a traditional malicious spyware operation not related to government mandate. The malware was not distributed via the official Google Play Store but rather through third-party platforms, likely using social engineering and phishing techniques.

b. Attack Analysis: Malicious Android spyware distributed via third-party

apps masquerading as legitimate Services, enabling covert Surveillance and data theft.

c. Indicators of Compromise (Iocs):

- Apps requesting excessive permissions.
- Unusual background activity and data transmission.
- Unauthorized microphone, camera, and GPS access.
- Battery drain and performance degradation.

d. Assessment: covert threat targeting personal privacy and sensitive mobile data.

05. Denail of Service Impact - cloudflare Tenant API Outage.

a. Introduction: A new version of the cloudflare Dashboard was released which contains a bug that triggered many more calls to the organizations endpoint, including retries in the event of failure.

b. Attack Analysis: A software bug that caused repeated, unnecessary calls to the Tenant Service API. The API calls were managed by a React useEffect hook, but we mistakenly included a problematic Object in its dependency array. Because this Object was recreated on every State or prop change, React treated it as "always new", causing the use effect to rerun each time. As a result, the API call executed many times during a single dashboard render instead of just once. This behavior coincided with a service update to the Tenant Service API, compounding instability and ultimately overwhelming the service, which then failed to recover.

c. Indicators of Compromise (Iocs):

- Sudden spike in API request volume.
- Repeated calls to organizations endpoint.
- Widespread 5xx server errors.
- Dashboard and API authorization failures.

d. Assessment: Availability-focused incident highlighting the risks of software misconfiguration and cascading failures.

05. RESULT:

The practical Successfully demonstrated how indicators of compromise play a critical role in early detection and incident prioritization. By correlating Iocs with real-world attack patterns, effective prevention and mitigation strategies were developed for each cyber threat category, enhancing organizational cyber resilience.

01. AIM: To analyze Signs of a potential security incident by monitoring System processes, startup entries, registry changes, and network activity using Sysinternals Suite tools.

02. OBJECTIVES:

- a. To understand how System-level monitoring helps detect Security incidents.
- b. To analyze running processes using Process Explorer.
- c. To identify Suspicious Startup entries using Autoruns.
- d. To monitor real-time registry and file systems activity using Process monitor.
- e. To analyze active network connections using Tcpview.
- f. To detect abnormal or malicious behavior in a system.

03. PROCEDURE:

Step 1: Setup.

- a. Download Sysinternals Suite from Microsoft website.
- b. Extract the zip file.
- c. Run tools as Administrator.

Step 2: Analyze Running Processes (Process Explorer)

- a. Open Process Explorer.
- b. Observe:
 - Running Processes hierarchy.
 - CPU and Memory usage.
- c. Check for:
 - Unknown or Suspicious process.
 - Unsigned executables.
- d. Right-click → Verify Image Signature.

Step 3: Check Startup Entries (Autoruns)

a. Open Autoruns

b. Go through tabs:

- Logon
- Services
- Scheduled Tasks.

c. Identify:

- Unusual Startup program.
- Unknown publishers.

d. Disable suspicious entries (do not delete directly)

Step 4: Monitor Registry & File Activity (Process Monitor)

a. Open Process Monitor.

b. Start capture (Ctrl+E).

c. Apply filters (e.g. process name)

d. Observe:

- Registry modifications.
- File access activity.

e. Look for:

- Frequent registry changes.
- Unauthorized file access.

Step 5: Analyze Network Activity (TCPView)

a. Open TCPView.

b. Monitor:

- Active connections.
- Local & remote IP addresses.

c. Identify:

- Unknown remote connections.

- Suspicious ports.

Step 6: Correlation

a. correlate findings from all tools:

- Suspicious process + Startup entry + Network Activity.

b. Identify potential indicators of compromise.

04. OBSERVATION:

a. All running processes appeared normal and belonged to trusted applications.

b. No Suspicious or Unknown Startup entries were found.

c. Registry activity was normal with ~~no~~ ~~usual~~ unusual modifications observed.

d. Network connections were only established with trusted and known services.

e. All major processes were properly signed and verified.

05. RESULT:

a. Successfully analyzed system behavior using Sysinternals tools.

b. Identified potential indicators of a security incident such as:

- Suspicious processes.
- Unauthorized startup entries.
- Abnormal registry activity.
- Unknown network connections.

c. concluded that continuous monitoring using Sysinternals tools is ineffective in detecting early signs of system compromise.

01. AIM: To Install, configure Wazuh for monitoring an Endpoint.

02. OBJECTIVE:

- a. To install and set up the Wazuh Security monitoring platform.
- b. To configure an endpoint (agent) for monitoring.
- c. To analyze system activity such as logs, processes, and security events.
- d. To understand real-time intrusion detection and alerting.

03. PROCEDURE:

Step 1: Install Wazuh Server:

- a. Download Wazuh using the official installation script.
- b. Run the installation command:

```
curl -sO https://packages.wazuh.com/4.7/wazuh-install.sh  
sudo bash wazuh-install.sh -a
```

- c. Wait for installation to complete (this installs manager + dashboard)

Step 2: Access Wazuh Dashboard

- a. Open browser and go to:

<https://<Server-ip>>

- b. Login using default credentials provided after installation.

Step 3: Install Wazuh Agent on Endpoint

- a. Download agent for your OS (Windows / Linux)
- b. Install using:

```
sudo apt install wazuh-agent
```

(for Linux)

Step 4: Configure Agent

a. Open agent config file:

```
Sudo nano /var/ossec/etc/ossec.conf
```

b. Add Server IP:

```
<Server>
```

```
<address> YOUR_SERVER_IP </address>
```

```
</Server>
```

Step 5: Register Agent:

a. On Server:

```
Sudo /var/ossec/bin/manage_agents
```

b. Add new agent → copy key.

c. On agent:

```
Sudo /var/ossec/bin/agent-auth -m <Server-ip>
```

Step 6: Start Services.

a. Start agent:

```
Sudo systemctl start Wazuh-agent
```

b. Enable:

```
Sudo systemctl enable Wazuh-agent
```

Step 7: Verify Monitoring.

a. Go to Wazuh dashboard

b. Check:

- Agents → Active status.

- Security events.

- logs and alerts.

04. OBSERVATION:

- a. Wazuh agent successfully connected to the server.
- b. Real-time logs from the endpoint were visible in the dashboard.
- c. Alerts were generated for system activities such as:
 - login attempts.
 - file changes.
 - running processes.
- d. No suspicious or malicious activity was detected during monitoring.

05. RESULT:

- a. Successfully installed and configured Wazuh for endpoint monitoring.
- b. Established communication between server and agent.
- c. Verified real-time monitoring and alert generation.
- d. System was functioning normally with no detected security threats.

01. AIM: To perform live incident response & data acquisition on a windows System using Helix

02. OBJECTIVE:

- a. To understand the concept of live incident Response.
- b. To perform real-time data acquisition on a running windows System.
- c. To use Helix for collecting volatile and non-volatile evidence.
- d. To analyze System activity Such as processes, network connections, and memory data.

03. PROCEDURE:

Step 1: Launch Helix

- a. Insert Helix CD/USB or mount ISO
- b. Open Helix interface in windows.
- c. Run as Administrator.

Step 2: Start Live Response.

- a. Select Live Response Option in Helix
- b. choose Windows Forensic Mode.

Step 3: collect Volatile Data

- a. capture currently running System data
 - running processes.
 - Active network connections.
 - logged-in users.
 - Open ports.

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Step 4: Memory Acquisition

- a. Use Helix tools to dump RAM:
 - Capture System memory.
 - Save memory image for analysis.

Step 5: collect System Information.

- a. Extract:
 - System date & time
 - OS details.
 - Installed programs.
 - Startup entries.

Step 6: Acquire Non-Volatile Data.

- a. collect
 - File System data.
 - Important logs.
 - Registry information.
- b. Save data to external Storage.

Step 7: Save and Document Evidence.

- a. Store all collected data Securely.
- b. Maintain chain of custody.
- c. Label files properly.

Teacher's Signature _____

04. OBSERVATION:

- a. Helix Successfully collected System data.
- b. Running processes and network connections were visible.
- c. Memory dump was created successfully.
- d. System logs and registry entries were extracted.
- e. No Suspicious activities was detected in the System (For Simple observation).

05. RESULT:

- a. Live incident response was performed Successfully using Helix
- b. Both volatile and non-volatile data were acquired.
- c. The collected data can be used for further forensic analysis.

01. AIM: Installation & Demonstration of Sawmill on Windows OS & Custom Report Generation.

02. OBJECTIVE:

- a. To install and configure Sawmill log analyzer on windows OS.
- b. To analyze log files using Sawmill.
- c. To generate a customized report from log data.
- d. To understand log analysis and reporting techniques.

03. PROCEDURE:

a. Installation of Sawmill on Windows:

Step 1: Download

- a. Go to official website.
- b. Download the windows Version (.exe installer)

Step 2: Install

- a. Double click the .exe file
- b. Follow Installation Steps:
 - Accept license agreement.
 - Choose installation directory.
 - click Install.

Step 3: Launch

- a. After Installation, open Sawmill from:
 - Desktop Shortcut OR
 - Start menu.

b. Initial Setup (Profile Creation)

When you open Sawmill.

1. Click "Add Profile".
2. Select log type (e.g., Web Server, Firewall, etc).
3. Provide:
 - log file location (e.g., c:\logs\access.log)
 - log format (Common log format / IIS / Apache)
4. Save the profile.

Demonstration: Processing logs

1. Select the created Profile
2. Click "Build Database"
3. Sawmill will:
 - Read log files.
 - Process entries.
 - Store data internally.
4. Once completed → click "View Reports"

Generating Reports

Sawmill automatically generates reports such as:

- a. Visitor Statistics.
- b. Page Views.
- c. Errors.
- d. Bandwidth usage.
- e. IP activity.

c. Creating a Custom Report.

Step 1: Go to Report Section

- click "Reports" → "Custom Reports".

Step 2: Define Report

- Choose fields:

- IP Address
- Data/Time
- URL Accessed
- Status code.

Step 3: Apply Filters.

Example:

- Filter by specific IP
- Filter by data Range
- Filter only by error logs (e.g. 404)

Step 4: Customize layout.

- Select: Table / chart format, Sorting (e.g. Most visits)

Step 5: Generate Report

- Click "Run Report".

d. Export Report:

- Export Formats available:

- PDF
- CSV
- HTML

Steps:

- click Export.
- Choose format.
- Save to desired format.

04 OBSERVATION:

- a. logs were Successfully uploaded and parsed.
- b. Sawmill identified:
 - Total requests
 - Top accessed URLs
 - Frequent IP Addresses.
- c. Graphical reports (charts & tables) were generated.
- d. Custom report Filtering worked correctly.

05. RESULT:

- a. Successfully installed Sawmill log analyzer on windows.
- b. Log file was analyzed efficiently.
- c. custom reports were generated using filters.
- d. Practical demonstrated real-time log analysis capability.

01. AIM: To monitor disk activity and identify processes where the disk read & write queue length exceeds 5 using windows monitoring tools.

02. OBJECTIVES:

- a. To monitor disk activity in window system.
- b. To identify processes where disk read/write queue length exceeds 5.
- c. To analyze system performance using Built-in windows monitoring tools.

03. PROCEDURE:

Step 1: Open Task Manager

- a. Press Ctrl + Shift + Esc
- b. Go to performance → Disk
- c. Observe:
 - Active time (%)
 - Read / write Speed.

Step 2: Open Resource Monitor.

- a. Press Win + R → type resmon → Enter
- b. Go to Disk tab
- c. check :
 - Disk Activity
 - Disk Queue length.
- d. Identify processes with high disk Usage.

Step 3: Open performance Monitor

- a. Press Win + R → type Perfmon → Enter
- b. Go to:
 - monitoring Tools → Performance Monitor
- c. click + (Add counter)
- d. Add:
 - PhysicalDisk → Avg. Disk Queue Length.
- e. click Ok.

Step 4: Generate Disk load (Testing)

- a. Copy large files or run:
 - Software installation.
 - Multiple file transfers.
- b. Observe changes in queue length.

Step 5: Identify Queue length > 5

- a. In Performance Monitor:
 - Look for spikes above 5
- b. In Resource Monitor:
 - check which process is causing high disk usage.

4. OBSERVATIONS.

- a. Disk activity increases when:
 - large file transfers occur
 - Multiple applications access disk simultaneously.
- b. When queue length > 5:
 - Disk becomes a bottleneck.
 - System performance slows down.
- c. Example Observation: chrome.exe, reaches 6-8 Queue length.

Q5. RESULT:

- a. Successfully monitored disk activity using windows tools.
- b. Identified processes causing high disk usage.
- c. Observed that queue length exceeding 5 indicates disk congestion & performance degradation.

01. AIM: To examine & analyze Files & Folders from local hard disk and network

02. OBJECTIVES:

- (a) To understand the process of forensic data Examination.
- (b) To explore file System Structure in storage devices.
- (c) To analyze file metadata and attributes.
- (d) To identify hidden and deleted files.
- (e) To compare data from local and network devices.

03. PROCEDURE:

Step 1: Launch FTK Imager on the System.

Step 2: click on File → Add Evidence Item.

Step 3: Select Drive

- choose:

- C: (recommended)

- or D:

- click Finish.

Step 4: Browse Files

- (a) Click the arrow(+) next to the drive

- (b) open folders like:

- Users

- Program files.

- Documents.

Step 5: View File Details (

click any file → look at bottom panel

You will see:

- File name • Size • created/ modified Time.
- Path • Type

Step 6: Find Hidden / System Files

(a) Look for:

- Greyed-out files
- System folders (like AppData)

04. OBSERVATION:

Sl. No	File / Folder Name	Location	File Type	Size	Remarks
1	Architectur2.jpeg	C:\Users\manos\Documents	Image	~51 KB	Successfully viewed with meta data.
2	NTFS Test	C:\Users\manos\Desktop	Folder	152 bytes	User-created folder with directory structure.
3	notes.txt	C:\Users\Manos\Desktop\NTFS Test	Text	0 KB	Empty text file viewed in hex format.
4	archive.zip	C:\Users\Manos\Documents	compressed	~240 MB	large compressed file observed.
5	CDTI Internship form.pdf	C:\Users\manos\Documents	PDF	~332 KB	Document file analyzed.
6	[root]	C:\Drive	System Directory	-	contains main File system Structure.
7	[orphan]	C:\Drive	System Area	-	Represents deleted / unlinked files.

Teacher's Signature _____

05. RESULT:

Files and folders from both the local hard disk and network drive were successfully examined using FTK Imager. The tool helped in viewing the file system structure, file properties, and metadata. It also helped in identifying hidden or deleted files where available. Thus, the practical was completed successfully.

01. AIM: To recover deleted files from given data source using autopsy.

02. OBJECTIVES:

- (a) To understand the concept of file recovery in digital Forensics.
- (b) To create and manage forensic cases.
- (c) To analyze Storage media using Autopsy.
- (d) To recover deleted & hidden files.
- (e) To interpret recovered data.

03. PROCEDURE:

1. Launched the Autopsy application.
2. created a new case and entered required case details.
3. Added a data Source by selecting Logical Fils.
4. Selected the folder Test Recovery containing deleted files.
5. configured ingest modules such as:
 - File Type Identification.
 - Recent Activity
 - Embedded File Extractor.
6. Start the analysis process.
7. After completion, navigated to
 - Data Sources → Test Recovery.
8. Viewed listed files & analyzed their metadata.
9. Identified files such as:
 - Sample.txt
 - Sample.docx
 - Image.jpeg.
10. Examined file details like:
 - File type

- MIME Type.

- Allocation Status.

11. Attempted recovery by extracting Selected files.

12. Documented findings in the observation table.

04. OBSERVATION:

S.No	File Name	File Type	Status	location	Remarks.
1.	Sample.txt	text	Allocated	/LogicalFileSet1/ Test_Recovery/ Sample.txt	File present, metadata visible.
2.	Sample.docx	Document	Allocated	/LogicalFileSet1/Te st_Recovery/Sampk -docx	File detected Successfully.
3.	Image.jpeg	Image	Allocated	/LogicalFileSet1/ Test_Recovery/ image.jpeg	Image data identified (JFIF/Exif)

05. RESULT:

Files from Selected Data Source were Successfully analyzed using Autopsy. The tool identified different file types & displayed their metadata. However, the files were found in allocated space, indicating they were not ~~capable~~ completely deleted. The analysis demonstrated how Autopsy can be used to examine & recover digital Evidence.